The University of Texas at Austin
In-Class Work 9

M339G Predictive Analytics

November 06, 2026

Regression Trees: Pruning.

Problem 9.1. Instead of growing a large tree and then pruning it, a modeler proposes to stop splitting as soon as no split decreases the RSS by more than some threshold. Which of the following is the main drawback of this proposal?

a. It always produces a tree that is larger than the pruned tree.
b. A split that looks worthless may be followed by a split that greatly reduces the RSS.
c. It requires computing the RSS on a validation set at every step.
d. It cannot be used when there is more than one predictor.
e. It always produces a tree with only one split.

Solution. The correct answer is **b.**
Stopping early is short-sighted: a split with a small decrease in RSS may lead to a later split with a large decrease in RSS. Growing a large tree first and then pruning it back avoids this problem.

- a. is false: stopping early tends to produce smaller trees, not larger ones.
- c. is false: the threshold rule uses only the training RSS.
- d. is false: the rule applies to any number of predictors.
- e. is false: the number of splits depends on the threshold.

Problem 9.2. Determine which of the following statements concerning cost complexity pruning of a regression tree is/are true.

I. As the tuning parameter α increases, the selected subtree has the same or fewer terminal nodes.
II. As the tuning parameter α increases, the training RSS of the selected subtree decreases.
III. When α is very large, the selected subtree consists of the root node only.

a. I only
b. II only
c. I and II only
d. I and III only
e. I, II, and III

Solution. The correct answer is **d.**
- I is true: the penalty $\alpha\ |T|$ grows with α, so larger values of α favor subtrees with fewer terminal nodes.
- II is false: a subtree with fewer terminal nodes fits the training data worse, so the training RSS of the selected subtree increases (or stays the same) as α increases.
- III is true: for large enough α, the penalty dominates the RSS, and the tree with a single terminal node is selected.

Problem 9.3. *Sample SRM Problems.*
Determine which of the following statements concerning decision tree pruning is/are true.

I. The recursive binary splitting method can lead to overfitting the data.
II. A tree with more splits tends to have lower variance.
III. When using the cost complexity pruning method, $\alpha = 0$ results in a very large tree.

a. None
b. I and II only
c. I and III only
d. II and III only
e. The correct answer is not given by a., b., c., or d.

Solution. The correct answer is **c.**
- I is true: the method optimizes with respect to the training set, so the tree may perform poorly on the test set.
- II is false: additional splits tend to increase variance, since they add to the complexity of the model.
- III is true: with $\alpha = 0$ there is no penalty on the number of terminal nodes, so only the training error is measured and the full tree is selected.

Problem 9.4. A regression tree is fit using cost complexity pruning. Consider the following statements:

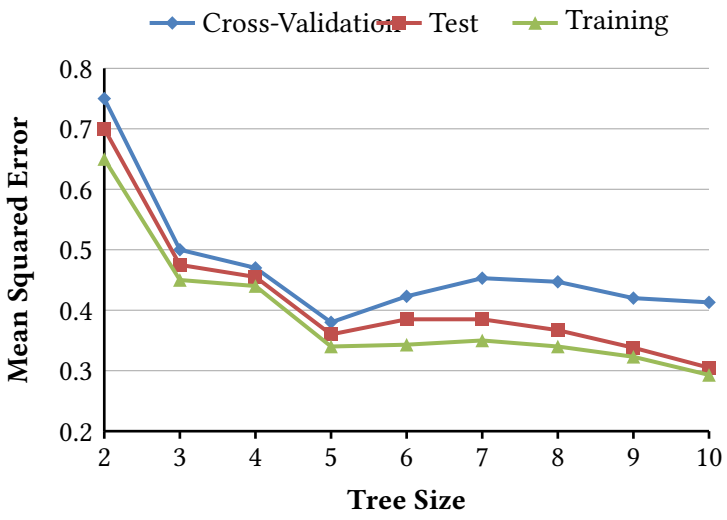
I. Cost complexity pruning compares every possible subtree of the large tree individually.
II. The tuning parameter α is typically chosen using K-fold cross-validation.
III. The final tree is obtained by pruning the large tree grown on the full training data, using the chosen value of α.

Determine which of the statements are true.

a. I only
b. II only
c. I and II only
d. II and III only
e. I, II, and III

Solution. The correct answer is **d.**
- I is false: comparing every subtree is computationally infeasible. Instead, cost complexity pruning considers a sequence of subtrees indexed by α.
- II is true: the value of α that minimizes the cross-validation error is selected.
- III is true: once α is chosen, the subtree corresponding to that α is taken from the tree grown on all of the training data.

Problem 9.5. You are using cost complexity pruning to determine the optimal regression tree, using mean squared error as the measure of prediction error. The graph below provides the mean squared error for three different approaches.

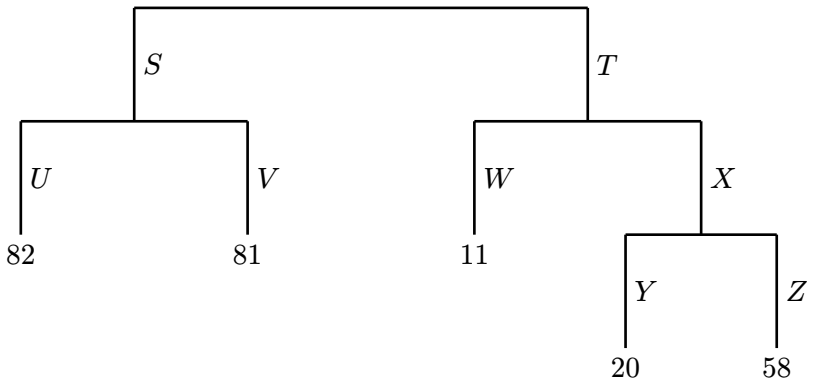


Determine the tree size of the optimal pruned tree.

a. 2
b. 5
c. 6
d. 9
e. 10

Solution. The correct answer is **b.**
Cost complexity pruning selects the tree size that minimizes the cross-validation MSE. The minimum of the cross-validation curve occurs at a tree of size five.

Problem 9.6. *Source: Exam MAS-II, Fall 2018.*
You are given the following unpruned decision tree, where the value at each terminal node is the residual sum of squares (RSS) at that node:



The table below gives the RSS at nodes S, T, and X if the tree was pruned at those nodes:

Node	RSS
S	251
T	209
X	86

The RSS for the null model is 486. You use the cost complexity pruning algorithm with the tuning parameter $\alpha = 9$ in order to evaluate the following pruning strategies:

I. No nodes pruned
II. Prune node S only
III. Prune node T only
IV. Prune node X only
V. Prune both nodes S and X

Determine which pruning strategy is selected.

a. I
b. II
c. III
d. IV
e. V

Solution. The correct answer is **d.**
Cost complexity pruning selects the subtree T that minimizes

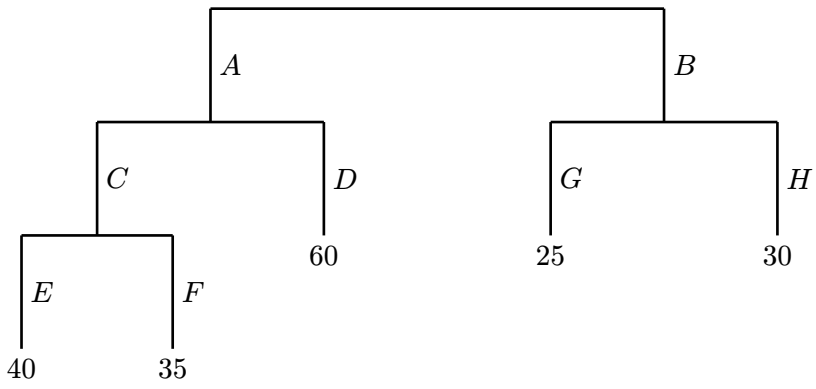
$$\text{RSS}(T) + \alpha\ |T|,$$

where $|T|$ is the number of terminal nodes and $\alpha = 9$.

- I: leaves U, V, W, Y, Z. $82 + 81 + 11 + 20 + 58 + 9(5) = 252 + 45 = 297$.
- II: leaves S, W, Y, Z. $251 + 11 + 20 + 58 + 9(4) = 340 + 36 = 376$.
- III: leaves U, V, T. $82 + 81 + 209 + 9(3) = 372 + 27 = 399$.
- IV: leaves U, V, W, X. $82 + 81 + 11 + 86 + 9(4) = 260 + 36 = 296$.
- V: leaves S, W, X. $251 + 11 + 86 + 9(3) = 348 + 27 = 375$.

The smallest value, 296, comes from pruning node X only. The null model RSS is not needed.

Problem 9.7. You are given the following unpruned decision tree, where the value at each terminal node is the residual sum of squares (RSS) at that node:



The table below gives the RSS at nodes A, B, and C if the tree was pruned at those nodes:

Node	RSS
A	190
B	80

C	95

The RSS for the null model is 520. You use the cost complexity pruning algorithm with the tuning parameter $\alpha = 22$ in order to evaluate the following pruning strategies:

I. No nodes pruned
II. Prune node C only
III. Prune node B only
IV. Prune both nodes B and C
V. Prune node A only

Determine which pruning strategy is selected.

a. I
b. II
c. III
d. IV
e. V

Solution. The correct answer is **b.**
Cost complexity pruning selects the subtree T that minimizes

$$\text{RSS}(T) + \alpha\ |T|,$$

where $|T|$ is the number of terminal nodes and $\alpha = 22$.

- I: leaves E, F, D, G, H. $40 + 35 + 60 + 25 + 30 + 22(5) = 190 + 110 = 300$.
- II: leaves C, D, G, H. $95 + 60 + 25 + 30 + 22(4) = 210 + 88 = 298$.
- III: leaves E, F, D, B. $40 + 35 + 60 + 80 + 22(4) = 215 + 88 = 303$.
- IV: leaves C, D, B. $95 + 60 + 80 + 22(3) = 235 + 66 = 301$.
- V: leaves A, G, H. $190 + 25 + 30 + 22(3) = 245 + 66 = 311$.

The smallest value, 298, comes from pruning node C only. The null model RSS is not needed.